

● Original Contribution

THE TREATMENT OF PRIMARY INTRAOCULAR MALIGNANCY

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This paper will summarize much of the information derived in an association between The Department of Radiation Oncology of Hahnemann University Hospital and the Oncology Service of Wills Eye Hospital of Thomas Jefferson University, a collaborative effort for the treatment of primary intraocular malignancies that has spanned the last dozen years. In that time we have treated malignant intraocular melanoma by radioactive eyeplaque brachytherapy and have begun to develop a similar program for treatment of recurring retinoblastoma. These experiences will be described.

Ocular melanoma, Uveal melanoma, Retinoblastoma, Brachytherapy, Eye plaques.

INTRODUCTION

Primary intraocular malignant disease is uncommon. In 1988, the most common malignant intraocular disease is metastatic from other primary sites. Of the primary intraocular malignant diseases, intraocular malignant melanoma comprises about 75% of all cases, retinoblastoma comprises about 20% and the remaining 5% includes exceedingly rare entities such as primary intraocular lymphoma and medulloepithelioma. Therefore, the remainder of this discussion will address only intraocular melanoma and retinoblastoma.

METHODS AND MATERIALS

Uveal melanoma

Although ophthalmologists have long been familiar with this disease entity, controversy still exists regarding optimal therapeutic modalities. Classically, enucleation of the affected eye has been the surgical therapy of choice. However, in certain circumstances, alternative surgical management by local resection,^{8,16,17,21,25,31} laser photocoagulation^{20,32} or cryopexy¹³ can be used. In fact, for very small lesions felt clinically to be melanomas, observation alone until the lesion has shown clinical growth has become a viable option.¹⁸

Enucleation as a modality for treating malignant mel-

anoma has been questioned by Zimmerman³³ and co-workers³⁴ who have suggested that prognosis following enucleation may be inferior to prognosis for untreated patients. Such a suggestion has led to the development of the so called "no touch" approach to enucleation,⁹ although it is not at all clear that tumor seeding is affected by manipulation of the human globe at time of removal.

Since the pioneering work of Stallard,^{28,29,31} radiation therapy as an alternative to enucleation has been investigated. Both external beam techniques^{4,5,10,11,24} and brachytherapeutic techniques^{2,3,6,14,15,16,23,29,31} have been examined and, by modern techniques, appear to yield grossly similar results when considering survival, local control and preservation of vision.

Modern approaches using external beam techniques have focussed on beams of heavy charged particles, taking advantage of the Bragg peak effect and extremely sharp penumbra of the beam to effectively treat the tumor to a high, homogeneous dose and relatively sparing nearby structures. Optimization of therapy requires exceptionally accurate tumor localization and demarcation, precise patient immobilization during therapy, verification of reproducibility on a daily basis, and complex treatment planning. This type of therapy is currently being performed at only two centers within the United States, at the Harvard Cyclotron and at the Berkeley laboratories facility.